

GATE CSE: 1. Discrete Maths - 1. Mathematical Logic

1. [GATE 1995 — 2 Mark]

If the proposition $\neg p \Rightarrow q$ is true, then the truth value of the proposition $\neg p \vee (p \Rightarrow q)$, where $\neg$ is negation, $\vee$ is inclusive or and $\Rightarrow$ is implication, is

- A. True
- B. Multiple valued
- C. False
- D. Cannot be determined

2. [GATE 1996 — 2 Mark]

Which one of the following is false? Read $\wedge$ as AND, $\vee$ as OR, $\sim$ as NOT, $\rightarrow$ as one way implication and $\leftrightarrow$ as two way implication.

A. $((x \rightarrow y) \wedge x) \rightarrow y$

B. $((\sim x \rightarrow y) \wedge (\sim x \rightarrow \sim y)) \rightarrow x$

C. $(x \rightarrow (x \vee y))$

D. $((x \vee y) \leftrightarrow (\sim x \rightarrow \sim y))$

3. [GATE 2000 — 2 Mark]

Let a, b, c, d be propositions. Assume that the equivalence $a \leftrightarrow (b \vee \neg b)$ and $b \leftrightarrow c$ hold. Then the truth-value of the formula $(a \wedge b) \rightarrow (a \wedge c) \vee d$ is always

- A. True
- B. False
- C. Same as the truth-value of b
- D. Same as the truth-value of d

4. [GATE 2001 — 1 Mark]

What is the converse of the following assertion? I stay only if you go

- A. I stay if you go
- B. If I stay then you go
- C. If you do not go then I do not stay
- D. If I do not stay then you go

5. [GATE 2001 — 1 Mark]

Consider two well-formed formulas in propositional logic: $F_1 : P \Rightarrow \neg P$ $F_2 : (P \Rightarrow \neg P) \vee (\neg P \Rightarrow P)$

Which of the following statements is correct?

- A. F_1 is satisfiable, F_2 is valid
- B. F_1 is unsatisfiable, F_2 is satisfiable
- C. F_1 is unsatisfiable, F_2 is valid
- D. F_1 and F_2 are both satisfiable

6. [GATE 2002 — 1 Mark]

"If X then Y unless Z" is represented by which of the following formulas in propositional logic? (" $\neg$ " is negation, " $\wedge$ " is conjunction, and " $\rightarrow$ " is implication)

A. $(X \wedge \neg Z) \rightarrow Y$

B. $(X \wedge Y) \rightarrow \neg Z$

C. $X \rightarrow (Y \wedge \neg Z)$

D. $(X \rightarrow Y) \wedge \neg Z$

7. [GATE 2003 — 2 Mark]

Which of the following is a valid first order formula? (Here α and β are first order formulae with x as their only free variable)

- A. $((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]) \Rightarrow (\forall x)[\alpha \Rightarrow \beta]$
- B. $(\forall x)[\alpha] \Rightarrow (\exists x)[\alpha \wedge \beta]$
- C. $(\forall x)[\alpha \vee \beta] \Rightarrow (\exists x)[\alpha] \Rightarrow (\forall x)[\alpha]$
- D. $(\forall x)[\alpha \Rightarrow \beta] \Rightarrow ((\forall x)[\alpha] \Rightarrow (\forall x)[\beta])$

8. [GATE 2003 — 2 Mark]

Consider the following formula α and its two interpretations I_1 and I_2 .

$$\alpha : (\forall x)[P_x \Leftrightarrow (\forall y)[Q_{xy} \Leftrightarrow \neg Q_{yy}]] \Rightarrow (\forall x)[\neg P_x]$$

I_1 : Domain: the set of natural numbers

P_x = 'x is a prime number'

Q_{xy} = 'y divides x'

I_2 : Same as I_1 except that P_x = 'x is a composite number.'

Which of the following statements is true?

- A. I_1 satisfies α , I_2 does not
- B. I_2 satisfies α , I_1 does not
- C. Neither I_2 nor I_1 satisfies α
- D. Both I_1 and I_2 satisfy α

9. [GATE 2003 — 2 Mark]

The following resolution rule is used in logic programming: Derive clause $(P \vee Q)$ from clauses $(P \vee R)$, $(Q \vee \neg R)$ Which of the following statements related to this rule is FALSE?

- A. $(P \vee R) \wedge (Q \vee \neg R) \Rightarrow (P \vee Q)$ is logically valid
- B. $(P \vee Q) \Rightarrow (P \vee R) \wedge (Q \vee \neg R)$ is logically valid
- C. $(P \vee Q)$ is satisfiable if and only if $(P \vee R) \wedge (Q \vee \neg R)$ is satisfiable
- D. $(P \vee Q) \Rightarrow FALSE$ if and only if both P and Q are unsatisfiable

10. [GATE 2004 — 1 Mark]

Identify the correct translation into logical notation of the following assertion. Some boys in the class are taller than all the girls

Note: Taller (x, y) is true if x is taller than y .

- A. $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
- B. $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
- C. $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$
- D. $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$

11. [GATE 2004 — 1 Mark]

Let $a(x, y)$, $b(x, y)$, and $c(x, y)$ be three statements with variables x and y chosen from some universe. Consider the following statement:

$$(\exists x)(\forall y)[(a(x, y) \wedge b(x, y)) \wedge \neg c(x, y)]$$

Which one of the following is its equivalent?

- A. $(\forall x)(\exists y)[(a(x, y) \vee b(x, y)) \rightarrow c(x, y)]$
- B. $(\exists x)(\forall y)[(a(x, y) \vee b(x, y)) \wedge \neg c(x, y)]$
- C. $\neg(\forall x)(\exists y)[(a(x, y) \wedge b(x, y)) \rightarrow c(x, y)]$
- D. $\neg(\forall x)(\exists y)[(a(x, y) \vee b(x, y)) \rightarrow c(x, y)]$

12. [GATE 2004 — 2 Mark]

Let p, q, r and s be four primitive statements. Consider the following arguments:

P: $[(\neg p \vee q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (\neg s \rightarrow q)$

Q: $[(\neg p \wedge q) \wedge [q \rightarrow (p \rightarrow r)]] \rightarrow \neg r$

R: $[[(q \wedge r) \rightarrow p] \wedge (\neg q \vee p)] \rightarrow r$ S: $[p \wedge (p \rightarrow r) \wedge (q \vee \neg r)] \rightarrow q$

Which of the above arguments are valid?

- A. P and Q only
- B. P and R only
- C. P and S only
- D. P, Q, R and S

13. [GATE 2004 — 2 Mark]

The following propositional statement is

$$(P \rightarrow (Q \vee R)) \rightarrow ((P \wedge Q) \rightarrow R)$$

- A. Satisfiable but not valid
- B. Valid
- C. A contradiction
- D. None of the above

14. [GATE 2005 — 2 Mark]

Let P , Q and R be three atomic propositional assertions. Let X denote $(P \vee Q) \rightarrow R$ and Y denote $(P \rightarrow R) \vee (Q \rightarrow R)$. Which one of the following is a tautology?

- A. $X \equiv Y$
- B. $X \rightarrow Y$
- C. $Y \rightarrow X$
- D. $\neg Y \rightarrow X$

15. [GATE 2005 — 2 Mark]

What is the first order predicate calculus statement equivalent to the following? Every teacher is liked by some student

- A. $\forall(x)[\text{teacher}(x) \rightarrow \exists(y)[\text{student}(y) \rightarrow \text{likes}(y, x)]]$
- B. $\forall(x)[\text{teacher}(x) \rightarrow \exists(y)[\text{student}(y) \wedge \text{likes}(y, x)]]$
- C. $\exists(y)\forall(x)[\text{teacher}(x) \rightarrow [\text{student}(y) \wedge \text{likes}(y, x)]]$
- D. $\forall(x)[\text{teacher}(x) \wedge \exists(y)[\text{student}(y) \rightarrow \text{likes}(y, x)]]$

16. [GATE 2005 — 2 Mark]

Let $P(x)$ and $Q(x)$ be arbitrary predicates. Which of the following statements is always TRUE?

- A. $(\forall x(P(x) \vee Q(x))) \Rightarrow ((\forall xP(x)) \vee (\forall xQ(x)))$
- B. $(\forall x(P(x) \Rightarrow Q(x))) \Rightarrow ((\forall xP(x)) \Rightarrow (\forall xQ(x)))$
- C. $(\forall x(P(x) \Rightarrow (\forall xQ(x)))) \Rightarrow (\forall x(P(x) \Rightarrow Q(x)))$
- D. $((\forall x(P(x)) \Leftrightarrow (\forall xQ(x))) \Rightarrow (\forall x(P(x) \Leftrightarrow Q(x)))$

17. [GATE 2006 — 2 Mark]

Consider the following first order logic formula in which R is a binary relation symbol.

$\forall x \forall y (R(x, y) \Rightarrow R(y, x))$ The formula is

- A. Satisfiable and valid
- B. Satisfiable and so is its negation
- C. Unsatisfiable but its negation is valid
- D. Satisfiable but its negation is unsatisfiable

18. [GATE 2006 — 2 Mark]

Which one of the first order predicate calculus statements given below correctly expresses the following English statement?

Tigers and lions attack if they are hungry or threatened.

- A. $\forall x[(\text{tiger}(x) \wedge \text{lion}(x)) \rightarrow \{(\text{hungry}(x) \vee \text{threatened}(x)) \rightarrow \text{attacks}(x)\}]]$
- B. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow \{(\text{hungry}(x) \vee \text{threatened}(x)) \wedge \text{attacks}(x)\}]]$
- C. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow \{(\text{attacks}(x) \rightarrow (\text{hungry}(x) \vee \text{Threatened}(x)))\}]]$
- D. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow \{(\text{hungry}(x) \vee \text{threatened}(x)) \rightarrow \text{attacks}(x)\}]]$

19. [GATE 2006 — 2 Mark]

Consider the following propositional statements:

$$P_1 : ((A \wedge B) \rightarrow C) \equiv ((A \rightarrow C) \wedge (B \rightarrow C))$$

$$P_2 : ((A \vee B) \rightarrow C) \equiv ((A \rightarrow C) \vee (B \rightarrow C))$$

Which one of the following is true?

- A. P_1 is a tautology, but not P_2
- B. P_2 is a tautology, but not P_1
- C. P_1 and P_2 are both tautologies
- D. Both P_1 and P_2 are not tautologies

20. [GATE 2006 — 2 Mark]

A logical binary relation $\odot$, is defined as follows:

A	B	$A \odot B$
TRUE	TRUE	TRUE
TRUE	FALSE	TRUE
FALSE	TRUE	FALSE
FALSE	FALSE	TRUE

Let $\sim$ be the unary negation (NOT) operator, with higher precedence, than $\odot$. Which one of the following is equivalent to $A \wedge B$?

- A. $(\sim A \odot B)$
- B. $\sim (A \odot \sim B)$
- C. $\sim (\sim A \odot \sim B)$
- D. $\sim (\sim A \odot B)$

21. [GATE 2007 — 2 Mark]

Let $\text{Graph}(x)$ be a predicate which denotes that x is a graph. Let $\text{Connected}(x)$ be a predicate which denotes that x is connected. Which of the following first order logic sentences DOES NOT represent the statement; "Not every graph is connected"?

- A. $\neg\forall x(\text{Graph}(x) \Rightarrow \text{Connected}(x))$
- B. $\exists x(\text{Graph}(x) \wedge \neg\text{Connected}(x))$
- C. $\neg\forall x(\neg\text{Graph}(x) \vee \text{Connected}(x))$
- D. $\forall x(\text{Graph}(x) \Rightarrow \neg\text{Connected}(x))$

22. [GATE 2007 — 2 Mark]

Which of the following is TRUE about formulae in Conjunctive Normal Form?

- A. For any formula, there is a truth assignment for which at least half the clauses evaluate to true.
- B. For any formula, there is a truth assignment for which all the clauses evaluate to true.
- C. There is a formula such that for each truth assignment at most one-fourth of the clauses evaluate to true.
- D. None of the above

23. [GATE 2007 — 2 Mark]

Which one of these first-order logic formulae is valid?

A. $\forall x(P(x) \Rightarrow Q(x)) \Rightarrow ((\forall xP(x)) \Rightarrow (\forall xQ(x)))$

B. $\exists x(P(x) \vee Q(x)) \Rightarrow ((\exists xP(x)) \Rightarrow (\exists xQ(x)))$

C. $\exists x(P(x) \wedge Q(x)) \Leftrightarrow ((\exists xP(x)) \wedge (\exists xQ(x)))$

D. $\forall x\exists yP(x, y) \Rightarrow \exists y\forall xP(x, y)$

24. [GATE 2008 — 1 Mark]

Let fsa and pda be two predicates such that $fsa(x)$ means x is a finite state automaton, and $pda(y)$ means, that y is a pushdown automaton. Let $equivalent$ be another predicate such that $equivalent(a, b)$ means a and b are equivalent. Which of the following first order logic statement represents the following:
Each finite state automaton has an equivalent pushdown automaton.

- A. $\forall x(fsa(x) \Rightarrow \exists y(pda(y) \wedge equivalent(x, y)))$
- B. $\sim \forall y(\exists x fsa(x) \Rightarrow pda(y) \wedge equivalent(x, y))$
- C. $\forall x \exists y(fsa(x) \wedge pda(y) \wedge equivalent(x, y))$
- D. $\forall x \exists y(fsa(y) \wedge pda(x) \wedge equivalent(x, y))$

25. [GATE 2008 — 2 Mark]

Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first order formula with x as a free variable, and β is a first order formula with no free variable.

- A. $[\beta \rightarrow (\exists x, \alpha(x))] \rightarrow [\forall x, \beta \rightarrow \alpha(x)]$
- B. $[\exists x, \beta \rightarrow \alpha(x)] \rightarrow [\beta \rightarrow (\forall x, \alpha(x))]$
- C. $[(\exists x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$
- D. $[(\forall x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$

26. [GATE 2008 — 2 Mark]

Which of the following is the negation of $[\forall x, \alpha \rightarrow (\exists y, \beta \rightarrow (\forall u, \exists v, \gamma))]$

- A. $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, \gamma))]$
- B. $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, \neg \gamma))]$
- C. $[\forall x, \neg \alpha \rightarrow (\exists y, \neg \beta \rightarrow (\forall u, \exists v, \neg \gamma))]$
- D. $[\exists x, \alpha \wedge (\forall y, \beta \wedge (\exists u, \forall v, \neg \gamma))]$

27. [GATE 2008 — 2 Mark]

P and Q are two propositions. which of the following logical expressions are equivalent?

1. $P \vee \sim Q$
2. $\sim (\sim P \wedge Q)$
3. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$
4. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge Q)$

- A. Only 1 and 2
- B. Only 1, 2 and 3
- C. Only 1, 2 and 4
- D. All of these

28. [GATE 2009 — 2 Mark]

Which one of the following is the most appropriate logical formula to represent the statement: "Gold and silver ornaments are precious"

The following notations are used:

$G(x)$: x is a gold ornament

$S(x)$: x is a silver ornament

$P(x)$: x is precious

A. $\forall x(P(x) \rightarrow (G(x) \wedge S(x)))$

B. $\forall x(G(x) \wedge (S(x) \rightarrow P(x)))$

C. $\exists x((G(x) \wedge S(x)) \rightarrow P(x))$

D. $\forall x((G(x) \vee S(x)) \rightarrow P(x))$

29. [GATE 2009 — 2 Mark]

The binary operation $\square$ is defined as follows:

P	Q	$P \square Q$
T	T	T
T	F	T
F	T	F
F	F	T

Which one of the following is equivalent to $P \vee Q$?

- A. $\neg Q \square \neg P$
- B. $P \square \neg Q$
- C. $\neg P \square Q$
- D. $\neg P \square \neg Q$

30. [GATE 2009 — 2 Mark]

Consider the following well-formed formulae:

I. $\neg\forall x(P(x))$

II. $\neg\exists x(P(x))$

III. $\neg\exists x(\neg P(x))$

IV. $\exists x(\neg P(x))$

Which of the above are equivalent?

A. I and III

B. I and IV

C. II and III

D. II and IV

31. [GATE 2010 — 2 Mark]

Suppose the predicate $F(x, y, t)$ is used to represent the statement that person x can fool person y at time t . Which one of the statements below expresses best the meaning of the formula $\forall x \exists y \exists t (\neg F(x, y, t))$?

- A. Everyone can fool some person at some time
- B. No one can fool everyone all the time
- C. Everyone cannot fool some person all the time
- D. No one can fool some person at some time

32. [GATE 2011 — 2 Mark]

Which one of the following options is CORRECT given three positive integers x , y and z , and a predicate

$$P(x) = \neg(x = 1) \wedge \forall y(\exists z(x = y * z) \Rightarrow (y = x) \vee (y = 1))$$

- A. $P(x)$ being true means that x is a prime number
- B. $P(x)$ being true means that x is a number other than 1
- C. $P(x)$ is always true irrespective of the value of x
- D. $P(x)$ being true means that x has exactly two factors other than 1 and x

33. [GATE 2012 — 1 Mark]

Consider the following logical inferences.

I_1 : If it rains then the cricket match will not be played.

The cricket match was played.

Inference: There was no rain.

I_2 : If it rains then the cricket match will not be played.

It did not rain.

Inference: The cricket match was played.

Which of the following is TRUE?

- A. Both I_1 and I_2 are correct inferences
- B. I_1 is correct but I_2 is not a correct inference
- C. I_1 is not correct but I_2 is a correct inference
- D. Both I_1 and I_2 are not correct inferences

34. [GATE 2012 — 1 Mark]

What is the correct translation of the following statement into mathematical logic?

”Some real numbers are rational”

- A. $\exists x(\text{real}(x) \vee \text{rational}(x))$
- B. $\forall x(\text{real}(x) \rightarrow \text{rational}(x))$
- C. $\exists x(\text{real}(x) \wedge \text{rational}(x))$
- D. $\exists x(\text{rational}(x) \rightarrow \text{real}(x))$

35. [GATE 2013 — 2 Mark]

What is the logical translation of the following statements?

”None of my friends are perfect”

A. $\exists x(F(x) \wedge \neg P(x))$

B. $\exists x(\neg F(x) \wedge P(x))$

C. $\exists x(\neg F(x) \wedge \neg P(x))$

D. $\neg \exists x(F(x) \wedge P(x))$

36. [GATE 2013 — 2 Mark]

Which one of the following is NOT logically equivalent to $\neg \exists x(\forall y(\alpha) \wedge \forall z(\beta))$

- A. $\forall x(\exists z(\neg\beta) \rightarrow \forall y(\alpha))$
- B. $\forall x(\forall z(\beta) \rightarrow \exists y(\neg\alpha))$
- C. $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg\beta))$
- D. $\forall x(\exists y(\neg\alpha) \vee \exists z(\neg\beta))$

37. [GATE 2014 — 1 Mark]

Consider the statement:

”Not all that glitters is gold”

Predicate $glitters(x)$ is true if x glitters and predicate $gold(x)$ is true if x is gold. Which one of the following logical formulae represents the above statement?

A. $\forall x : glitters(x) \Rightarrow \neg gold(x)$

B. $\forall x : gold(x) \Rightarrow glitters(x)$

C. $\exists x : gold(x) \wedge \neg glitters(x)$

D. $\exists x : glitters(x) \wedge \neg gold(x)$

38. [GATE 2014 — 2 Mark]

Which one of the following propositional logic formulas is TRUE when exactly two of p , q , and r are TRUE?

- A. $((p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
- B. $(\sim (p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
- C. $((p \rightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
- D. $(\sim (p \leftrightarrow q) \wedge r) \wedge (p \wedge q \wedge \sim r)$

39. [GATE 2014 — 2 Mark]

Which one of the following Boolean expressions is NOT a tautology?

A. $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$

B. $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$

C. $(a \wedge b \wedge c) \rightarrow (c \vee a)$

D. $a \rightarrow (b \rightarrow a)$

40. [GATE 2014 — 1 Mark]

Consider the following statements:

P: Good mobile phones are not cheap

Q: Cheap mobile phones are not good

L: P implies Q

M: Q implies P

N: P is equivalent to Q

Which one of the following about L, M, and N is
CORRECT?

A. Only L is TRUE

B. Only M is TRUE

C. Only N is TRUE

D. L, M and N are TRUE

41. [GATE 2014 — 2 Mark]

The CORRECT formula for the sentence, "not all rainy days are cold" is

- A. $\forall d(\text{Rainy}(d) \wedge \sim \text{Cold}(d))$
- B. $\forall d(\sim \text{Rainy}(d) \rightarrow \text{Cold}(d))$
- C. $\exists d(\sim \text{Rainy}(d) \rightarrow \text{Cold}(d))$
- D. $\exists d(\text{Rainy}(d) \wedge \sim \text{Cold}(d))$

42. [GATE 2015 — 1 Mark]

Which one of the following is Not equivalent to $p \leftrightarrow q$?

A. $(\sim p \vee q) \wedge (p \vee \sim q)$

B. $(\sim p \vee q) \wedge (q \rightarrow p)$

C. $(\sim p \wedge q) \vee (p \wedge \sim q)$

D. $(\sim p \wedge \sim q) \vee (p \wedge q)$

43. [GATE 2015 — 2 Mark]

The binary operator $\oplus$ is defined by the following truth table.

p	q	$p \neq q$
0	0	0
0	1	1
1	0	1
1	1	0

Which one of the following is true about the binary operator $\oplus$?

- A. Both commutative and associative
- B. Commutative but not associative
- C. Not commutative but associative
- D. Neither commutative nor associative

44. [GATE 2015 — 1 Mark]

Consider the following two statements:

S_1 : If a candidate is known to be corrupt, then he will not be elected.

S_2 : If a candidate is kind, he will be elected.

Which one of the following statements follows from S_1 and S_2 as per sound inference rules of logic?

- A. If a person is known to be corrupt, he is kind
- B. If a person is not known to be corrupt, he is not kind
- C. If a person is kind, he is not known to be corrupt
- D. If a person is not kind, he is not known to be corrupt

45. [GATE 2015 — 2 Mark]

Which one of the following well formed formulae is a tautology?

A. $\forall x \exists y R(x, y) \leftrightarrow \exists y \forall x R(x, y)$

B. $(\forall x [\exists y R(x, y) \rightarrow S(x, y)]) \rightarrow \forall x \exists y S(x, y)$

C. $[\forall x \exists y (P(x, y) \rightarrow R(x, y))] \leftrightarrow [\forall x \exists y (\neg P(x, y) \vee R(x, y))]$

D. $\forall x \forall y P(x, y) \rightarrow \forall x \forall y P(y, x)$

46. [GATE 2015 — 1 Mark]

In a room there are only two types of people, namely Type 1 and Type 2. Type 1 people always tell the truth and Type 2 people always lie. You give a fair coin to a person in that room, without knowing which type he is from and tell him to toss it and hide the result from you till you ask for it. Upon asking, the person replies the following:

“The result of the toss is head if and only if I am telling the truth.”
Which of the following options is correct?

- A. The result is head
- B. The result is tail
- C. If the person is of Type 2, then the result is tail
- D. If the person is of Type 1, then the result is tail

47. [GATE 2016 — 1 Mark]

Let p, q, r, s represent the following propositions.

p : $x \in \{8, 9, 10, 11, 12\}$

q : x is a composite number

r : x is a perfect square

s : x is a prime number

The integer $x \geq 2$ which satisfies $\neg((p \Rightarrow q) \wedge (\sim r \vee \sim s))$ is _____

Answer: _____

48. [GATE 2016 — 1 Mark]

Consider the following expressions:

(i) False

(ii) Q

(iii) True

(iv) $P \vee Q$

(v) $\neg Q \vee P$

The number of expressions given above that are logically implied by $P \wedge (P \Rightarrow Q)$ is _____

Answer: _____

49. [GATE 2016 — 2 Mark]

Which one of the following well-formed formulae in predicate calculus is NOT valid?

- A. $(\forall x p(x) \Rightarrow \forall x q(x)) \Rightarrow (\exists x \neg p(x) \vee \forall x q(x))$
- B. $(\exists x p(x) \vee \exists x q(x)) \Rightarrow \exists x (p(x) \vee q(x))$
- C. $(\exists x (p(x) \wedge q(x))) \Rightarrow (\exists x p(x) \wedge \exists x q(x))$
- D. $\forall x (p(x) \vee q(x)) \Rightarrow (\forall x p(x) \vee \forall x q(x))$

50. [GATE 2017 — 1 Mark]

Consider the first-order logic sentence $\forall x \exists y R(x, y)$. Assuming non-empty logical domains, which of the sentences below are implied by F ?

I. $\exists y \forall x R(x, y)$

II. $\exists y \forall x R(x, y)$

III. $\forall y \exists x R(x, y)$

IV. $\exists x \forall y \neg R(x, y)$

A. IV only

B. I and IV only

C. II only

D. II and III only

51. [GATE 2017 — 1 Mark]

The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?

I. $p \Rightarrow q$

II. $q \Rightarrow p$

III. $(\neg q) \vee p$

IV. $(\neg p) \vee q$

A. I only

B. I and IV only

C. II only

D. II and III only

52. [GATE 2026 — 1 Mark]

Let p, q , and r be propositions and the expression $(p \Rightarrow q) \Rightarrow r$ be a contradiction. Then, the expression $(r \Rightarrow p) \Rightarrow \neg q$ is

- A. A tautology
- B. A contradiction
- C. Always TRUE when p is FALSE
- D. Always TRUE when q is TRUE

53. [GATE 2026 — 1 Mark]

Let p, q, r denote the statements “It is raining”, “It is cold”, and “It is pleasant”, respectively. Then the statement “It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold” is represented by

- A. $(\neg p \wedge r) \wedge (\neg r \Rightarrow (p \wedge q))$
- B. $(\neg p \wedge r) \wedge ((p \wedge q) \Rightarrow \neg r)$
- C. $(\neg p \wedge r) \vee ((p \wedge q) \Rightarrow \neg r)$
- D. $(\neg p \wedge r) \vee (\neg r \Rightarrow (p \wedge q))$

54. [GATE 2019 — 2 Mark]

Consider this first order predicate formula

$$\forall x[(\forall z; z \mid x \Rightarrow (z = x \vee z = 1)) \Rightarrow \exists u(w > x) \vee (w = z) \vee (z = 1)]$$

Here $a \mid b$ denotes that “ a divides b ”, where a and b are integers. Consider the following sets:

S_1 : 1, 2, 3, ..., 100

S_2 : Set of all positive integers

S_3 : Set of all integers

Which of the above sets satisfy φ ?

- A. S_1 and S_3
- B. S_2 and S_3
- C. S_1, S_2 and S_3
- D. S_1 and S_2

55. [GATE 2020 — 2 Mark]

Which one of the following predicate formulae is NOT logically valid? Note that W is a predicate formula without any free occurrence of x .

- A. $\exists x(p(x) \wedge W) \equiv \exists xp(x) \wedge W$
- B. $\forall x(p(x) \Rightarrow W) \equiv \forall xp(x) \Rightarrow W$
- C. $\exists x(p(x) \Rightarrow W) \equiv \forall xp(x) \Rightarrow W$
- D. $\forall x(p(x) \vee W) \equiv \forall xp(x) \vee W$

56. [GATE 2021 — 1 Mark]

Let p and q be two propositions. Consider the following formulae in propositional logic.

$$S_1 : (\neg p \vee q) \Rightarrow q$$

$$S_2 : q \Rightarrow (p \vee q)$$

Which one of the following choices is correct?

- A. Neither S_1 nor S_2 is a tautology
- B. Both S_1 and S_2 are tautologies
- C. S_1 is a tautology but S_2 is not a tautology
- D. S_1 is not a tautology but S_2 is a tautology

57. [GATE 2021 — 1 Mark]

Choose the correct choice(s) regarding the following propositional logic assertion S :

$$S : ((P \wedge Q) \Rightarrow R) \Rightarrow ((P \wedge Q) \Rightarrow (Q \Rightarrow R))$$

- A. S is a tautology
- B. S is neither a tautology nor a contradiction
- C. The antecedent of S is logically equivalent to consequent of S
- D. S is a contradiction

58. [GATE 2023 — 1 Mark]

Geetha has a conjecture about integers, which is of the form $\forall x(P(x) \Rightarrow \exists yQ(x, y))$, where P is a statement about integers, and Q is a statement about pairs of integers. Which of the following (one or more) option(s) imply Geetha's conjecture?

- A. $\exists x(P(x) \wedge \forall yQ(x, y))$
- B. $\forall x\forall yQ(x, y)$
- C. $\exists y\forall x(P(x) \Rightarrow Q(x, y))$
- D. $\exists x(P(x) \wedge \exists yQ(x, y))$

59. [GATE 2024 — 1 Mark]

Let p and q be the following propositions:

p : Fail grade can be given.

q : Student scores more than 50% marks.

Consider the statement “Fail grade cannot be given when student scores more than 50% marks.”

Which one of the following is the CORRECT representation of the above statement in propositional logic?

A. $q \Rightarrow p$

B. $q \Rightarrow \neg p$

C. $p \Rightarrow q$

D. $\neg p \Rightarrow \neg q$